

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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Course Code/ Name: DSA0601 /Data Handling and Visualization

Slot C | CO1 – AT2 (R Only)

Q1: Color Scales for Data Representation [R]

<i>Employee</i>	<i>Performance Score</i>
<i>E1</i>	62
<i>E2</i>	78
<i>E3</i>	85
<i>E4</i>	70
<i>E5</i>	91
<i>E6</i>	55

1. Write an R program to create a bar chart using a sequential color scale based on Performance Score.

```
library(ggplot2)

employee_data <- data.frame(
  Employee = c("E1", "E2", "E3", "E4", "E5", "E6"),
  Performance = c(62, 78, 85, 70, 91, 55)
)

ggplot(employee_data,
  aes(x = Employee, y = Performance, fill = Performance)) +
  geom_col() +
  scale_fill_gradient(low = "lightblue", high = "darkblue") +
  labs(title = "Employee Performance Scores",
    x = "Employee", y = "Performance Score") +
  theme_minimal()
```

2. Identify the employees with the highest and lowest performance scores from the visualization.

Highest: E5 with a score of 91. Lowest: E6 with a score of 55.

3. Explain why a sequential color scale is appropriate.

A sequential color scale is appropriate because Performance Score is numerical and ordered. Lighter shades can represent lower values and darker shades can represent higher values.

4. Suggest a color palette suitable for color-blind users.

Use a color-blind-friendly sequential palette such as Viridis.

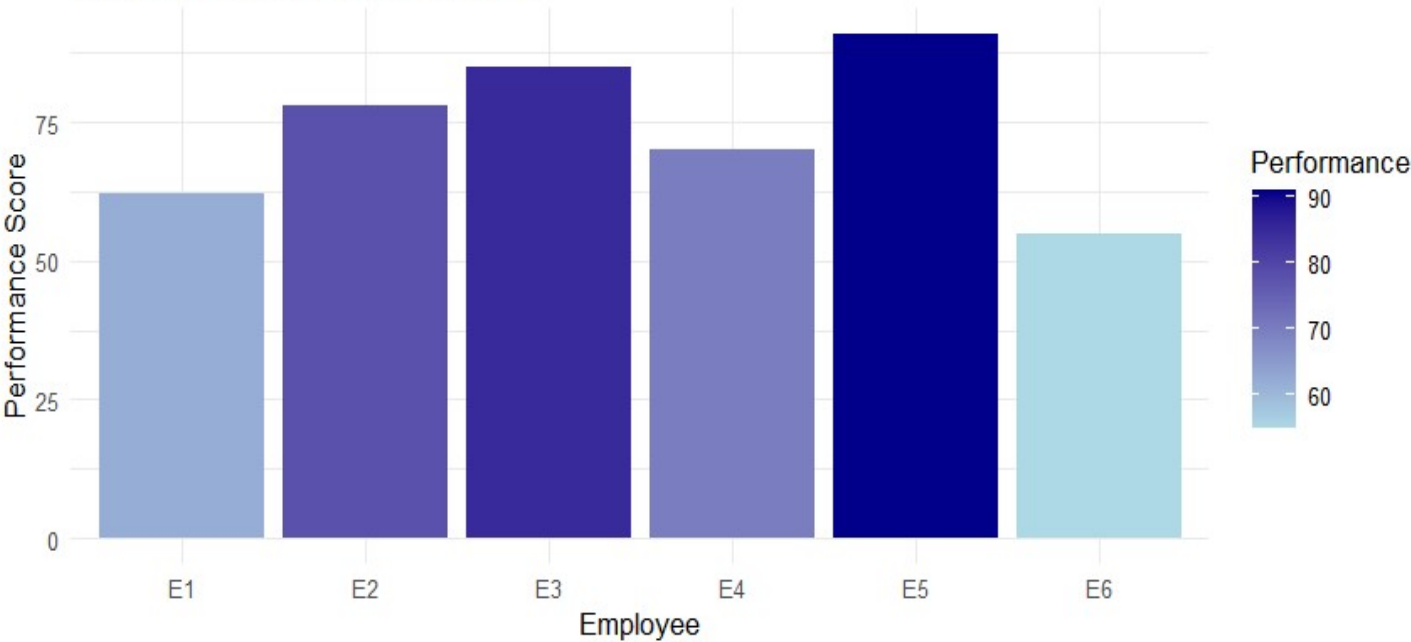
```
scale_fill_viridis_c()
```

5. Explain the difference between sequential and categorical color scales.

A sequential scale represents ordered numerical values from low to high. A categorical scale represents distinct groups with no numerical ordering. Example: Performance Score → sequential; Department → categorical.

Expected output:

Employee Performance Scores



Q2: Highlighting with Color [R]

<i>Expense Category</i>	<i>Monthly Spend (Rs)</i>
<i>Rent</i>	15000
<i>Groceries</i>	8000
<i>Travel</i>	4000
<i>Utilities</i>	3000
<i>Entertainment</i>	2500

1. Write an R program to create a bar chart highlighting the expense category with the highest monthly spend.

```
library(ggplot2)

expense_data <- data.frame(
  Category = c("Rent", "Groceries", "Travel", "Utilities", "Entertainment"),
  Spend = c(15000, 8000, 4000, 3000, 2500)
)

expense_data$Highlight <- expense_data$Spend == max(expense_data$Spend)

ggplot(expense_data,
  aes(x = Category, y = Spend, fill = Highlight)) +
  geom_col() +
  scale_fill_manual(values = c("FALSE" = "grey70",
    "TRUE" = "red")) +
  labs(title = "Monthly Expenses",
    x = "Expense Category", y = "Monthly Spend (Rs)") +
  theme_minimal() +
  theme(legend.position = "none")
```

2. Explain how highlighting improves data interpretation.

Highlighting draws attention to an important value and helps the viewer identify the key category quickly.

3. Identify the highest-spending category.

Rent – Rs 15,000.

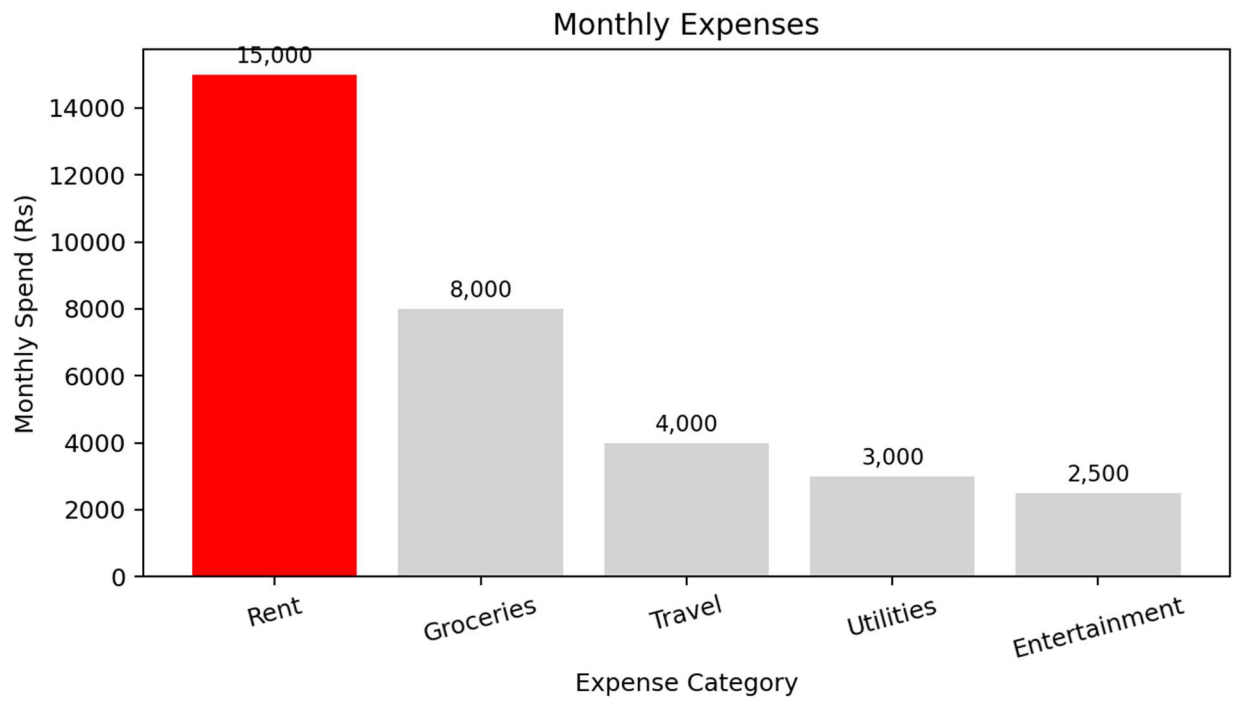
4. Discuss how excessive use of colors can affect interpretation.

Too many colors can make the chart confusing, distract the viewer, and make it difficult to identify the main information.

5. Suggest one alternative highlighting technique other than color.

Use a different pattern, thicker outline, or a data label to emphasize the highest-spending bar.

Expected output:



Q3: Stacked Bar Charts and Proportions [R]

<i>Quarter</i>	<i>Brand P (%)</i>	<i>Brand Q (%)</i>	<i>Brand R (%)</i>
<i>Q1</i>	35	40	25
<i>Q2</i>	38	36	26
<i>Q3</i>	42	33	25
<i>Q4</i>	45	30	25

1. Write an R program to create a stacked bar chart showing each brand's market share (%) by Quarter.

```
library(ggplot2)
library(tidyr)

market_data <- data.frame(
  Quarter = c("Q1", "Q2", "Q3", "Q4"),
  Brand_P = c(35, 38, 42, 45),
  Brand_Q = c(40, 36, 33, 30),
  Brand_R = c(25, 26, 25, 25)
)

long_data <- pivot_longer(
  market_data,
  cols = c(Brand_P, Brand_Q, Brand_R),
  names_to = "Brand",
  values_to = "MarketShare"
)

ggplot(long_data,
  aes(x = Quarter, y = MarketShare, fill = Brand)) +
  geom_col() +
  labs(title = "Market Share by Quarter",
    x = "Quarter", y = "Market Share (%)") +
  theme_minimal()
```

2. Modify the program to create a 100% (proportional) stacked bar chart.

```
ggplot(long_data,
  aes(x = Quarter, y = MarketShare, fill = Brand)) +
  geom_col(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  labs(title = "Proportional Market Share by Quarter",
    x = "Quarter", y = "Proportion") +
  theme_minimal()
```

3. Compare what each version communicates about the brands' relative performance.

The regular stacked bar chart shows the total and individual contributions. The 100% stacked bar chart normalizes every quarter to 100%, making the relative proportions easier to compare.

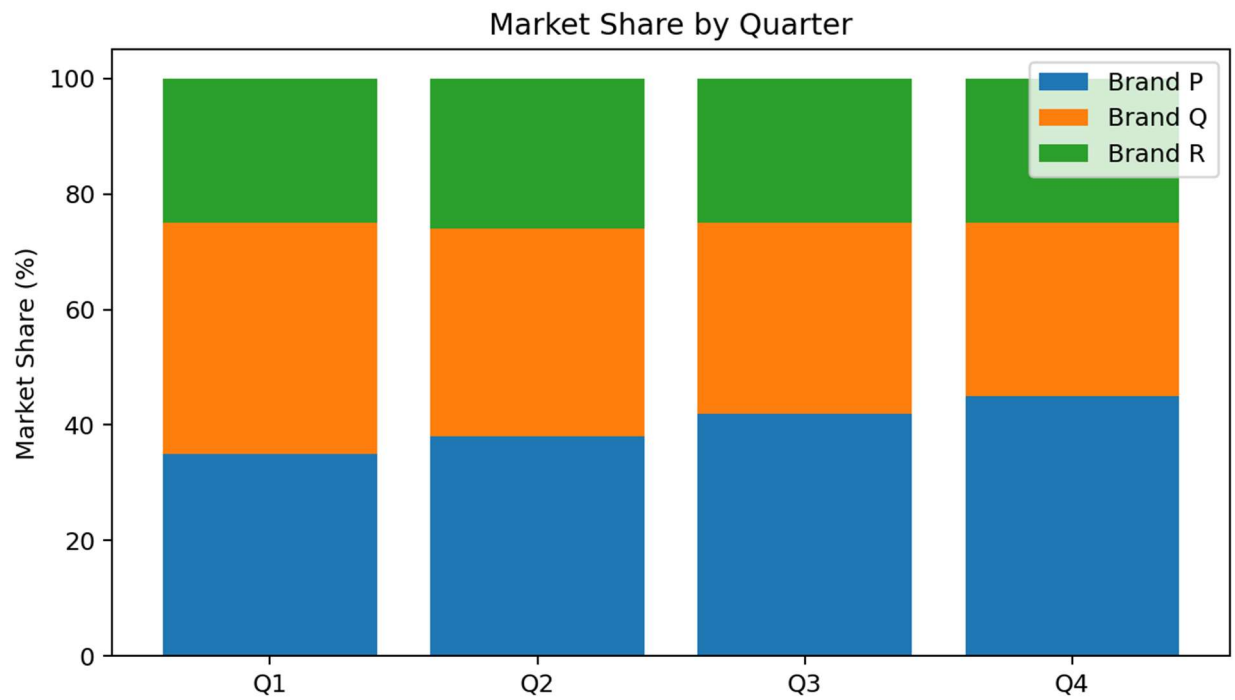
4. Identify the brand with a consistently rising trend.

Brand P: 35% → 38% → 42% → 45%.

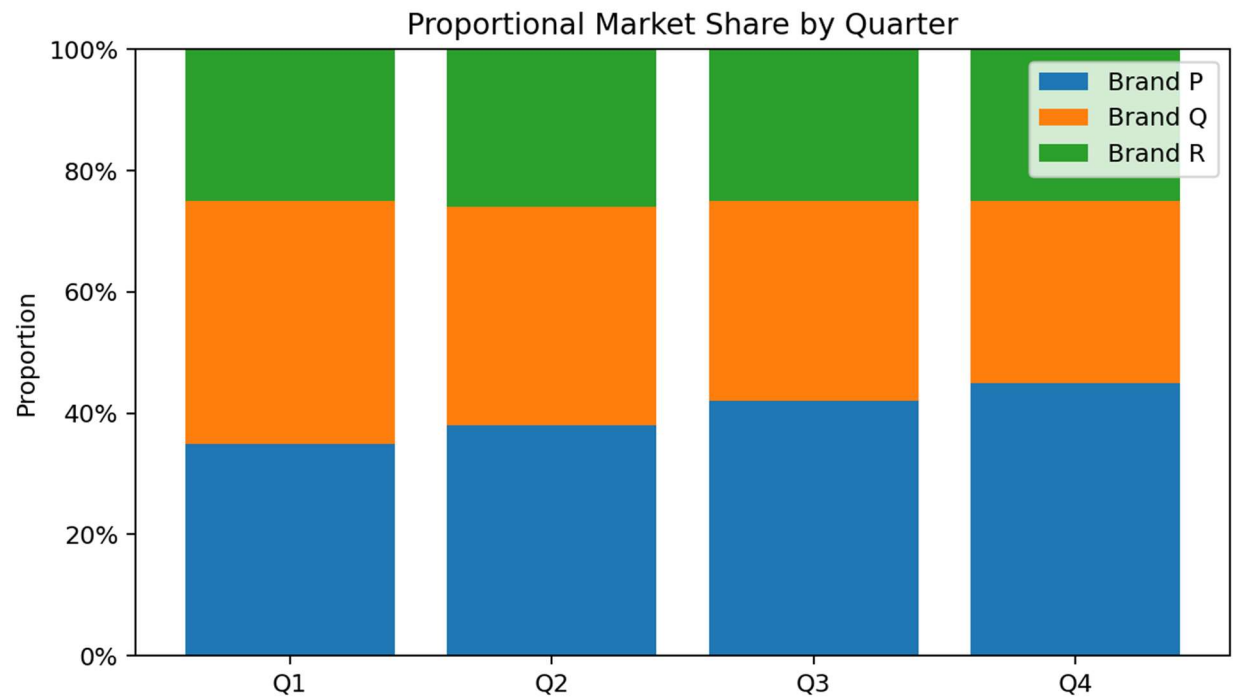
5. Explain a limitation of stacked bar charts when comparing individual segment sizes.

Individual segments can be difficult to compare because segments may start from different baselines.

Expected output – Regular stacked bar:



Expected output – 100% proportional stacked bar:



Q4: Integrated Visualization [R]

<i>Country</i>	<i>GDP (Rs Tn)</i>	<i>Literacy (%)</i>	<i>Population (Mn)</i>	<i>Continent</i>
<i>C1</i>	12.5	92	140	<i>Asia</i>
<i>C2</i>	8.2	88	65	<i>Europe</i>
<i>C3</i>	21.0	95	330	<i>N. America</i>
<i>C4</i>	3.1	75	210	<i>S. America</i>
<i>C5</i>	5.6	80	95	<i>Africa</i>

1. Write an R program to create a bubble chart using appropriate aesthetic mappings.

```
library(ggplot2)

country_data <- data.frame(
  Country = c("C1", "C2", "C3", "C4", "C5"),
  GDP = c(12.5, 8.2, 21.0, 3.1, 5.6),
  Literacy = c(92, 88, 95, 75, 80),
  Population = c(140, 65, 330, 210, 95),
  Continent = c("Asia", "Europe", "N. America",
    "S. America", "Africa")
)

ggplot(country_data,
  aes(x = GDP, y = Literacy,
    size = Population, color = Continent)) +
  geom_point(alpha = 0.8) +
  geom_text(aes(label = Country), vjust = -1, size = 4) +
  labs(title = "GDP, Literacy and Population by Country",
    x = "GDP (Rs Trillion)", y = "Literacy Rate (%)",
    size = "Population (Mn)", color = "Continent") +
  theme_minimal()
```

2. Select suitable aesthetics for GDP, Literacy Rate, Population, and Continent.

GDP → x-axis; Literacy Rate → y-axis; Population → bubble size; Continent → color; Country → label.

3. Interpret the relationship between GDP and Literacy Rate.

In this small dataset, higher GDP generally occurs with higher literacy rates. This describes an association in these observations and does not establish causation.

4. Identify the categorical and numerical variables.

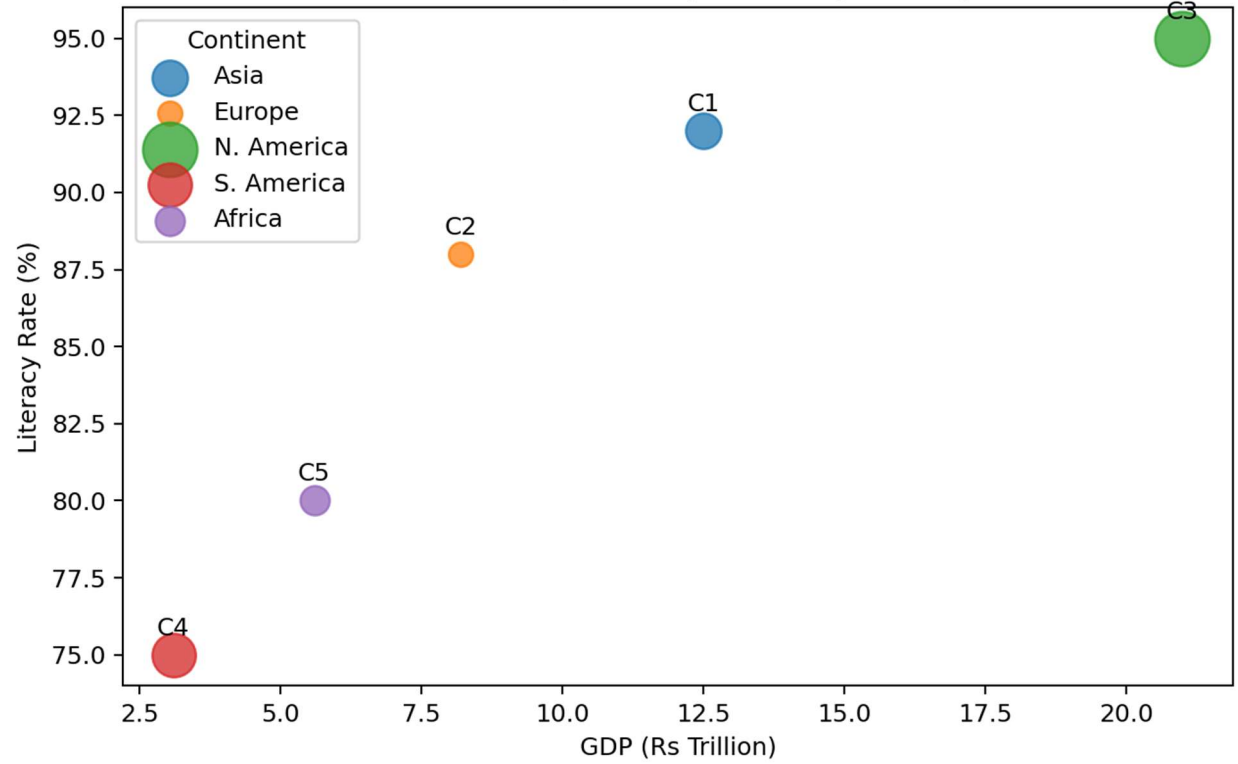
Numerical: GDP, Literacy Rate, Population. Categorical: Country and Continent.

5. Recommend one improvement to enhance the visualization's readability.

Use clear labels and a color-blind-friendly palette while keeping the number of visual elements limited.

Expected output:

GDP, Literacy and Population by Country



Q5: Mapping Data onto Aesthetics [R]

<i>Item</i>	<i>Orders</i>	<i>Rating (/5)</i>	<i>Category</i>
<i>I1</i>	320	4.2	<i>Electronics</i>
<i>I2</i>	540	3.8	<i>Grocery</i>
<i>I3</i>	210	4.6	<i>Fashion</i>
<i>I4</i>	690	4.0	<i>Grocery</i>
<i>I5</i>	150	4.8	<i>Fashion</i>
<i>I6</i>	410	3.5	<i>Electronics</i>

1. Write an R program (using ggplot2) to create a scatter plot with Orders on the x-axis and Rating on the y-axis.

```
library(ggplot2)

item_data <- data.frame(
  Item = c("I1", "I2", "I3", "I4", "I5", "I6"),
  Orders = c(320, 540, 210, 690, 150, 410),
  Rating = c(4.2, 3.8, 4.6, 4.0, 4.8, 3.5),
  Category = c("Electronics", "Grocery", "Fashion",
               "Grocery", "Fashion", "Electronics")
)

ggplot(item_data,
  aes(x = Orders, y = Rating,
      color = Category, size = Orders)) +
  geom_point(alpha = 0.8) +
  geom_text(aes(label = Item), vjust = -1, size = 4) +
  labs(title = "Orders vs Rating",
       x = "Number of Orders", y = "Rating (/5)",
       color = "Category", size = "Orders") +
  theme_minimal()
```

2. Map Category using an appropriate aesthetic (e.g. color) and Orders using another suitable aesthetic (e.g. size).

Category → color. Orders → x-axis and bubble size. Rating → y-axis.

3. Interpret the relationship between Orders and Rating.

The data does not show a strong positive relationship between Orders and Rating. I4 has the highest orders (690) but a 4.0 rating, while I5 has the highest rating (4.8) but only 150 orders.

4. Identify the type of scale for each variable (nominal, ordinal, continuous).

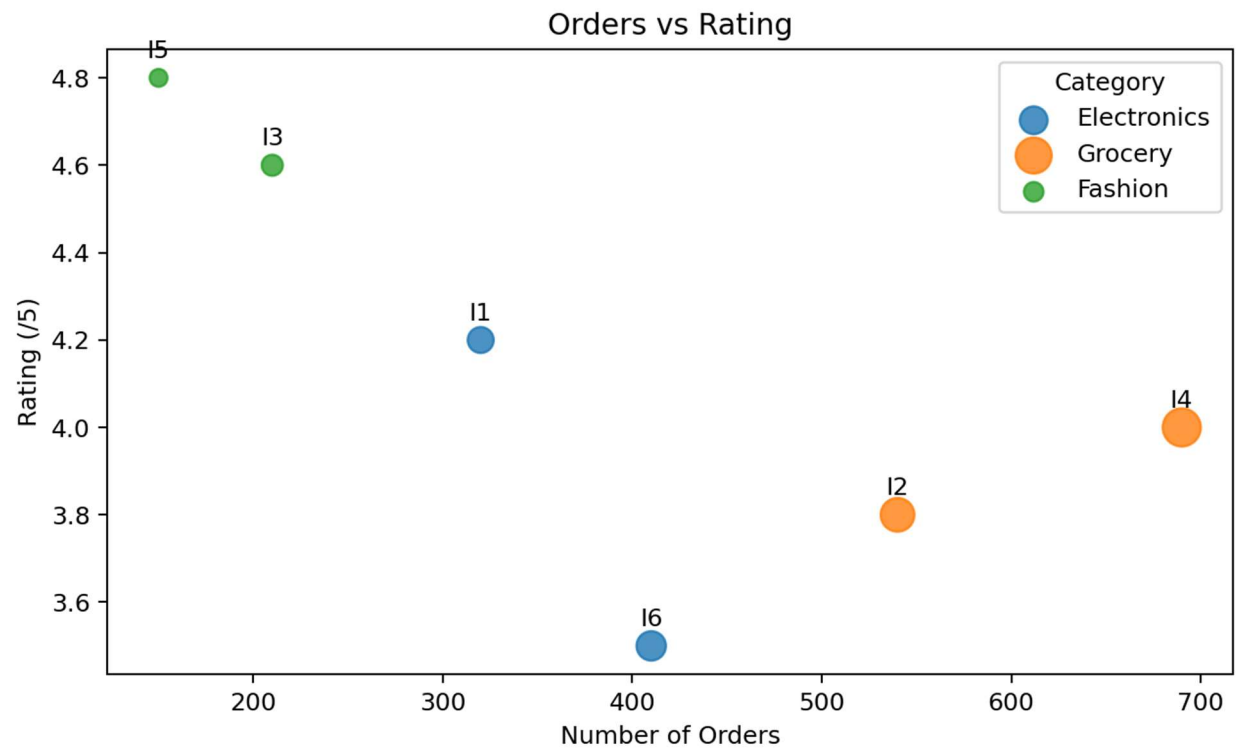
<i>Variable</i>	<i>Scale Type</i>
<i>Item</i>	<i>Nominal (identifier/category label)</i>
<i>Orders</i>	<i>Continuous / ratio numerical</i>
<i>Rating</i>	<i>Ordinal</i>
<i>Category</i>	<i>Nominal</i>

5. Suggest one alternative visualization that better represents the same data.

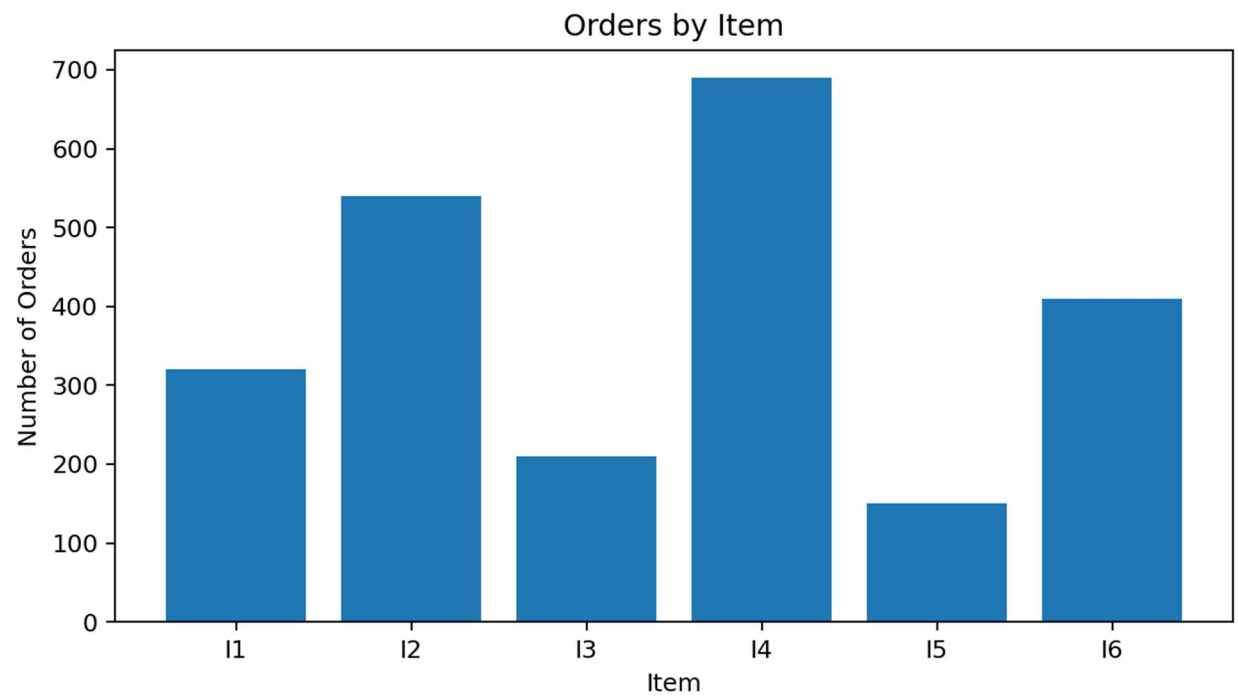
A bar chart of Orders by Item, with color used to distinguish Category, directly represents the order counts. Rating can be shown separately with labels or a second chart.

```
ggplot(item_data,
  aes(x = Item, y = Orders, fill = Category)) +
  geom_col() +
  labs(title = "Orders by Item",
       x = "Item", y = "Number of Orders") +
  theme_minimal()
```

Expected output – Scatter plot:



Expected output – Alternative bar chart:



Quick Answer Reference

<i>Question</i>	<i>Key Answer</i>
<i>Q1 highest performance</i>	<i>E5 = 91</i>
<i>Q1 lowest performance</i>	<i>E6 = 55</i>
<i>Q2 highest spending</i>	<i>Rent = Rs 15,000</i>
<i>Q3 consistently rising brand</i>	<i>Brand P</i>
<i>Q3 Brand P trend</i>	<i>35% → 38% → 42% → 45%</i>
<i>Q4 highest GDP</i>	<i>C3 = Rs 21.0 Tn</i>
<i>Q4 highest literacy</i>	<i>C3 = 95%</i>
<i>Q4 largest population</i>	<i>C3 = 330 Mn</i>
<i>Q5 highest orders</i>	<i>I4 = 690</i>
<i>Q5 highest rating</i>	<i>I5 = 4.8</i>
<i>Q5 lowest rating</i>	<i>I6 = 3.5</i>